

Roadmap to Become a Data Analyst (Free Resources Guide)

This guide provides a practical roadmap for students who want to become Data Analysts. It focuses on the essential skills, free learning resources, and realistic time estimates to help you build strong analytical skills and a portfolio.

Phase 1 – Excel Fundamentals (2–3 Weeks)

- Learn Excel basics: formulas, pivot tables, data cleaning, and dashboards.
- Understand functions like VLOOKUP, INDEX-MATCH, IF statements.
- Practice analyzing simple datasets.

Free Resources	Where to Practice
YouTube: ExcellisFun	Kaggle datasets
Microsoft Excel tutorials	Practice with real CSV datasets
Coursera (audit mode)	Create small dashboards

Phase 2 – SQL for Data Analysis (3–4 Weeks)

- Learn how databases work.
- Practice SELECT queries, filtering, joins, group by, aggregations.
- Learn to extract insights from data.

Free Resources	Where to Practice
SQLBolt.com	SQLBolt exercises
Mode SQL Tutorial	LeetCode SQL problems
YouTube: freeCodeCamp SQL course	HackerRank SQL

Phase 3 – Python for Data Analysis (4–6 Weeks)

- Learn Python basics.
- Understand pandas, numpy, and data cleaning.
- Perform data analysis and manipulation.

Free Resources	Where to Practice
freeCodeCamp Python course	Kaggle notebooks
YouTube: Corey Schafer	Google Colab projects
Python pandas documentation	Analyze Kaggle datasets

Phase 4 – Statistics & Data Analysis Concepts (2–3 Weeks)

- Understand mean, median, variance, and probability.

- Learn hypothesis testing and distributions.
- Apply statistical thinking when analyzing datasets.

Phase 5 – Data Visualization (2–3 Weeks)

- Learn how to communicate insights visually.
- Create dashboards and reports.
- Tools to learn: Power BI or Tableau.

Free Resources	Where to Practice
Microsoft Learn Power BI	Power BI sample datasets
Tableau Public tutorials	Create dashboards
YouTube: Alex The Analyst	Public datasets

Phase 6 – Build Projects (4–6 Weeks)

- Analyze real-world datasets (sales data, COVID datasets, finance data).
- Build dashboards and publish results.
- Upload projects to GitHub and document insights.

Important Tips for Aspiring Data Analysts

- Focus on practical projects rather than collecting certificates.
- Practice SQL and data cleaning regularly.
- Build a portfolio with at least 3–5 projects.
- Use Kaggle to explore datasets and practice analysis.
- Learn storytelling with data – explaining insights clearly is crucial.

Estimated Timeline: If you study consistently for 2–3 hours daily, you can build strong data analysis skills within 4–6 months.